Caroline Daniel

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Post doctorate researcher in community ecology

WORK EXPERIENCE

Post doctorate on species coexistence and soil legacy effects in grassland ecosystems **July 2024 - Today**
Location: Terrestrial ecology group, Bern University
Supervised by: Madhav P. THAKUR

PhD student on plant species coexistence, impact of nutrients and pathogens on plant communities **August 2019 - May 2024**
Location: Community ecology group, Bern University
Supervised by: Eric ALLAN & Oscar GODOY

Publications:

- ***Daniel, C.**, Allan, E., Saiz, H., & Godoy, O. (2024). Fast–slow traits predict competition network structure and its response to resources and enemies. Ecology Letters, 27(4), e14425.*
- ***Daniel, C.**, Godoy, O., Pichon, N. A., Gheno, M., & Allan, E. (2025). Stably coexisting communities deliver higher ecosystem multifunctionality. bioRxiv, 2025-11. (preprint)*
- ***Daniel C.**, Godoy O., Allan E., Plant species route to coexistence: the impact of functional traits, resource and enemies, In prep.*

Master's internship: remote sensing of sun induced chlorophyll fluorescence and reflectance to assess photosynthetic activity at leaf scale **February 2018 - August 2018**
Location: ESE, University Paris-Saclay
Supervised by: Kamel SOUDANI

Master's internship: the protection of the alpine snowbell, *Soldanella alpina* **June 2017 - August 2017**
Location: ESE, University Paris-Saclay
Supervised by: Peter STREB

- **Publication:** *Bidel, L. P., Meyer, S., Talhouët, A. C., Baudin, X., **Daniel, C.**, Cazals, G., & Streb, P. (2020). Epidermal UVA screening capacity measured in situ as an indicator of light acclimation state of leaves of a very plastic alpine plant Soldanella alpina L. Plant Physiology and Biochemistry, 151, 10-20.*

Master's internship: the phylogeography of the african mangoose, *Atilax paludinosus* **February 2017 - May 2017**
Location: ISYEB, Muséum National d'Histoire Naturelle, Paris
Supervised by: Géraldine VERON

- **Publication:** *Veron, G., **Daniel, C.**, Pagani, P., Do Linh San, E., Kitchener, A. C., & Hassanin, A. (2023). A tale of two African mongooses (Carnivora: Herpestidae): differing genetic diversity and geographical structure across a continent. Mammalian Biology, 103(1), 37-52.*

EDUCATION

University of Bern **2024**

PhD in community ecology, defended on the 24th of May 2024

University Paris-Saclay **2018**

Double Master's degree:

- 'Biology, Ecology and Evolution' (BEE) with a specialization in 'Biodiversity and ecosystem functioning, and ecological engineering'
- 'Biology and biotechnology'

University Paris-Saclay **2016**

Bachelor's degree in 'Biology of organisms and evolution' (BOE)

Ecole Nationale Supérieure de la Nature et du Paysage, Blois **2013**

First year in Landscape engineering

OUTREACH

BES 2025 **December 2025**

Higher-order interactions modify the effects of global changes on community assembly (poster)

Workshop: Linking higher-order and indirect interactions to ecosystem stability **June 2025**

Organisation committee leader

BES 2024 **December 2024**

More strongly coexisting communities deliver higher ecosystem multifunctionality (session talk)

Swiss global change day **April 2023**

How do changes in agricultural treatments impact biodiversity? (poster)

Biology 23 **February 2023**

Dynamic changes in plant competition, effect of nitrogen and pathogens on competition networks (session talk)

Autumn School EcoSum "Thinking community ecology in the Anthropocene" **November 2022**

How do resource and enemies impact plant competition? (poster)

INTECOL22 **August 2022**

Dynamic changes in plant species competition: a network approach in the Pathogen Nitrogen Diversity (PaNDiv) experiment (session talk)

Workshop: The consequences of indirect and higher order interactions for ecosystem functioning **May 2021**

Organisation committee member

BES 2019 **December 2019**

Impact of resources and enemies on plant species coexistence (poster)

Other workshop participations:

- Critical transitions, early-warning signs, and coexistence theory (June 2022)
- Causal Inference for Biologist Graduate, Structural Equation Modelling (November 2019)

TEACHING EXPERIENCE

Courses

- R buddy club (Bachelor/Master)

Role: Organisation committee, sessions presentation
Semesters: Autumn semester 2022 – Autumn semester 2025 (every semester, ongoing)
Workload: 5 hours per month

- Terrestrial ecology seminar (Bachelor/Master)

Role: session presentation (once per semester) and weekly seminar discussions
Semesters: Autumn semester 2024, Autumn semester 2025 (ongoing)
Workload: 1.5 hours per week

- Scientific methods in Ecology and Evolution (Bachelor)

Role: session presentation
Semesters: Spring semester 2025 (ongoing)
Workload: 2 days per semester

- Journal Club in plant diversity (Master)

Role: session presentation (once per semester) and weekly seminar discussions
Semesters: Autumn semester 2019 – Spring semester 2024 (every semester)
Workload: 2 hour per week

- Advanced ecology course (Master)

Role: session presentation and field project supervision
Semesters: Spring semester 2020 – Spring semester 2023 (every Spring semester)
Workload: 2 days of presentation and six weeks of field project supervision per semester

- Biodiversity Sustainability (Bachelor)

Role: session presentation and field project supervision
Semesters: Spring semester 2020 – Spring semester 2023 (every Spring semester)
Workload: 2 days per semester (one day of presentation, one day of field project supervision)

Student supervision

Master student:

- Mathis GHENO *(April-August 2022)*

Bachelor students:

- Valentina MENDOZA *(Spring semester 2023)*
- Jérôme BECHTOLD *(Autumn semester 2022)*
- Eva BURGUNDER *(Autumn semester 2021)*
- Vera ALESSANDRELLO *(Spring semester 2020)*

Research practicals:

- Silja ELLER *(Spring semester 2023)*
- Tatjana WEY *(Autumn semester 2022)*
- Melanie CHRISTEN *(Autumn semester 2021)*

SKILLS

Coding: R, Matlab/Python, QGIS

Languages: French (native), English (advanced), Spanish (good), German (basic)